Prescriptive Analytics

End Term Assignment

Question 1: Optimal Product Mix for Neighborhood Patisserie

Part 1: Linear Programming Model Formulation

Based on the problem description, we need to determine the mix of Seeded and Multigrain loaves to maximize profit within the resource limits

1. **Decision Variables:** Let x_1 be the number of Seeded Loaves to produce. Let x_2 be the number of Multigrain Loaves to produce.
2. **Objective Function:** We want to maximize total profit.

$$\text{Maximize } Z = 25x_1 + 32x_2$$

(Where 25 is the profit for Seeded and 32 is the profit for Multigrain)

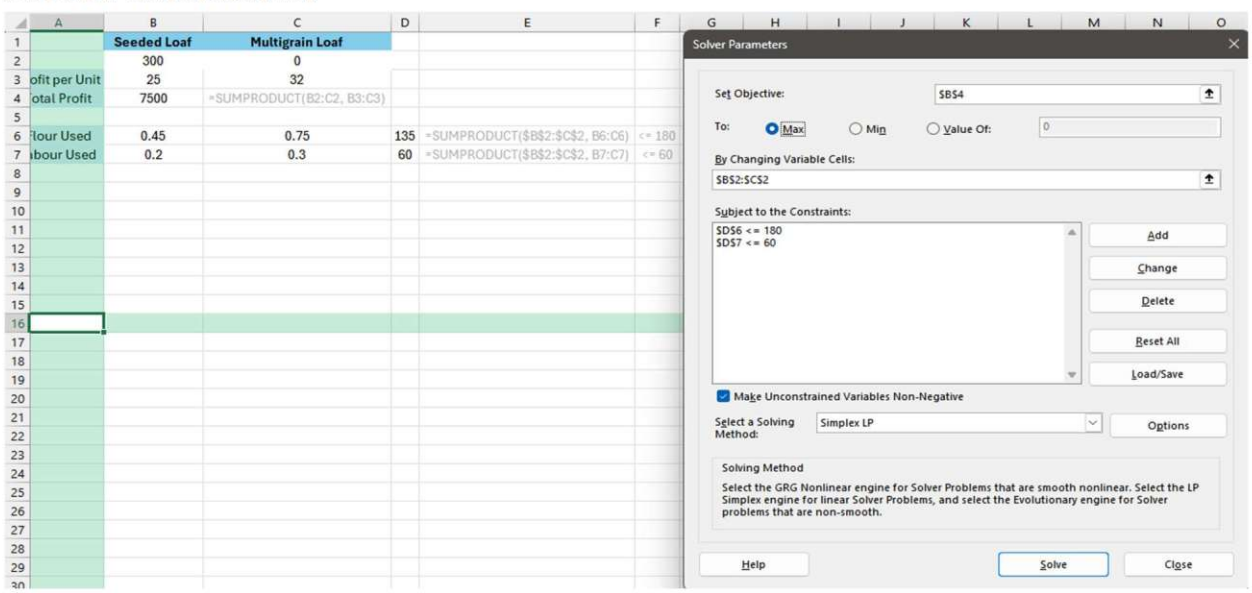
3. **Constraints:**
   - **Flour Constraint:** The total flour used must not exceed 180 kg.

     $$0.45x_1 + 0.75x_2 \leq 180$$
   - **Labour Constraint:** The total labour hours used must not exceed 60 hours.

     $$0.20x_1 + 0.30x_2 \leq 60$$
   - **Non-negativity Constraint:** You cannot produce negative loaves.

     $$x_1 \geq 0, x_2 \geq 0$$

Part 2: Excel Solver



Microsoft Excel 16.0 Sensitivity Report
Worksheet: [Q1_b.xlsx]Sheet1
Report Created: 2025-12-02 5:17:38 PM

Variable Cells

Cell	Name	Final Value	Reduced Cost	Objective Coefficient	Allowable Increase	Allowable Decrease
$B$2	Seeded Loaf	300	0	25	1E+30	3.666666667
$C$2	Multigrain Loaf	0	-5.5	32	5.5	1E+30

Constraints

Cell	Name	Final Value	Shadow Price	Constraint R.H. Side	Allowable Increase	Allowable Decrease
$D$6	Flour Used	135	0	180	1E+30	45
$D$7	Labour Used	60	125	60	20	60

Microsoft Excel 16.0 Answer Report
Worksheet: [Q1_b.xlsx]Sheet1
Report Created: 2025-12-02 5:17:38 PM
Result: Solver found a solution. All Constraints and optimality conditions are satisfied.
Solver Engine
Engine: Simplex LP
Solution Time: 0.016 Seconds.
Iterations: 1 Subproblems: 0
Solver Options
Max Time Unlimited, Iterations Unlimited, Precision 0.000001, Use Automatic Scaling
Max Subproblems Unlimited, Max Integer Sols Unlimited, Integer Tolerance 1%, Assume NonNegative

Objective Cell (Max)

Cell	Name	Original Value	Final Value
$B$4	Total Profit Seeded Loaf	7500	7500

Variable Cells

Cell	Name	Original Value	Final Value	Integer
$B$2	Seeded Loaf	300	300	Contin
$C$2	Multigrain Loaf	0	0	Contin

Constraints

Cell	Name	Cell Value	Formula	Status	Slack
$D$6	Flour Used	135	$D$6<=180	Not Binding	45
$D$7	Labour Used	60	$D$7<=60	Binding	0

Part 3: Solution & Interpretation

Optimal Production Plan

The linear programming model recommends the following daily production mix to maximize profit:

- **Seeded Loaf:** 300 units
- **Multigrain Loaf:** 0 units
- **Maximum Total Profit:** ₹7,500

Interpretation of Shadow Prices

The shadow prices (found in the Sensitivity Report) represent the marginal value of one additional unit of a resource.

1. **Shadow Price for Flour = ₹0**
   - **Meaning:** The shadow price is zero because Flour is a **non-binding** constraint. You have 180 kg available but only used 135 kg in the optimal plan. Since you already have 45 kg of leftover flour, acquiring more flour will not increase your profit.
2. **Shadow Price for Labour = ₹125**
   - **Meaning:** Labour is a **binding** constraint (you used all 60 hours). The shadow price indicates that if you could acquire **1 additional hour** of labour (increasing availability from 60 to 61 hours), your total profit would increase by **₹125**.
   - **Managerial Insight:** This suggests the bakery should be willing to pay up to ₹125 per hour for overtime or temporary labor, as it would still lead to a net increase in profit.

Question 2: Managerial Insights & Sensitivity Analysis

Profit of Seeded Loaf (₹)	Seeded Units (x1)	Multigrain Units (x2)	Total Profit (Z)
20	0	200	6,400
25 (Current)	300	0	7,500
30	300	0	9,000
35	300	0	10,500

1. Sensitivity of the Production Mix:

The optimal production plan is highly sensitive to the profitability of the Seeded Loaf. Specifically, we observe a complete shift in production strategy when the profit drops from ₹25 to ₹20.

2. The "Profit per Hour" Driver:

Since Labour is the binding constraint (the bottleneck resource), the decision of which bread to bake depends on which product generates the most profit *per hour of labour*, not just per loaf.

- **Multigrain Loaf Efficiency:** Fixed at ₹106.67 per hour (₹32 profit / 0.30 hours).
- **Seeded Loaf Efficiency at ₹20:** Drops to ₹100 per hour (₹20 profit / 0.20 hours).
- **Seeded Loaf Efficiency at ₹25:** Stands at ₹125 per hour (₹25 profit / 0.20 hours).

3. Strategic Recommendation:

- **When to Produce Seeded Loaves:** As long as the profit for Seeded Loaves remains above approximately **₹21.34**, they yield a higher return on labour hours than Multigrain Loaves, and the bakery should dedicate all capacity to them (300 units).
- **When to Switch:** If the profit margin for Seeded Loaves falls below this threshold (as seen in the ₹20 scenario), the bakery should switch production entirely to Multigrain Loaves (200 units) to maximize returns.

Question 3: Predictive Modeling for Subscription Renewal

a) Explain the Naïve Bayes classification approach and its independence assumption.

Naïve Bayes is a probabilistic machine learning algorithm based on Bayes' Theorem. It predicts the class of a data point (in this case, whether a customer will **Renew**) by calculating the probability of that class given the observed features (Usage Level and Support Tickets).

Why it assumes Conditional Independence: It is called "Naïve" because it makes a strong simplifying assumption: **Conditional Independence**. It assumes that the effect of one feature (e.g., Usage Level) on the class is completely independent of the other features (e.g., Support Tickets), *given* the class label.

Formally, this means the algorithm assumes:

$$P(\text{Usage}, \text{Tickets}|\text{Renew}) = P(\text{Usage}|\text{Renew}) \times P(\text{Tickets}|\text{Renew})$$

Why make this assumption? Without this assumption, we would need a massive dataset to observe every possible combination of features (e.g., specifically "High Usage" AND "Many Tickets" together) to calculate probabilities. By assuming independence, the algorithm can look at each feature separately, which drastically simplifies the computation and reduces the amount of training data required.

b) Illustrate how Naïve Bayes computes the posterior probability.

To determine if a customer will renew (Class = Yes), the algorithm calculates the **Posterior Probability** proportional to the product of the **Prior Probability** and the individual **Likelihoods** of the features[2].

For a customer with **High Usage** and **Many Tickets**, the calculation for "Renew = Yes" would look like this:

Posterior (Yes) ∝ Prior (Yes) × Likelihood (High Usage | Yes) × Likelihood (Many Tickets | Yes)

The algorithm performs the same calculation for "Renew = No". It then compares the two resulting scores. The class with the higher score is the predicted outcome.

Naive Bayes Classification

$$P(Y/X) = \frac{P(X/Y) * P(Y)}{P(X)}$$

c) Influence of the conditional independence assumption when features are correlated.

The problem states that there *is* a correlation: "High Usage customers tend to raise More Tickets". This violates the independence assumption.

The "Double Counting" Effect: When two features are correlated, they provide redundant information. For example, if "High Usage" strongly suggests Renewal, and "High Usage" *causes* "Many Tickets," then "Many Tickets" is just repeating the same signal.

Because Naïve Bayes treats them as independent, it multiplies their probabilities together as if they were two separate, distinct pieces of evidence. This leads to **"double counting"** the evidence.

Impact on Prediction: This usually results in **overconfidence**. The model pushes the predicted probabilities closer to 0 or 1 than they should be (e.g., predicting a 99% chance of renewal when the true probability might be only 80%).

d) Practical consequences and performance in real business datasets.

Two Practical Consequences:

1. **poor Probability Estimation (Calibration):** While the model might get the final category right, the probability scores it outputs is often unreliable. A business cannot trust the specific number (e.g., "there is a 92% risk of churn") for financial risk planning because the correlation inflates the score.
2. **Skewed Feature Importance:** In some cases, correlated features can dominate the decision-making process, drowning out weaker but non-correlated signals that might actually be important for specific edge cases.

Does it still perform well? Yes. Despite the violation of independence, Naïve Bayes is often surprisingly accurate in real-world business scenarios.

Reason: Classification depends only on which side of the decision boundary the point falls (e.g., Is Probability > 50%?). Even if the correlation makes the probability "wrong" (e.g., calculating 0.95 instead of 0.75), the **ranking order** usually remains correct. As long as the "Yes" score is still higher than the "No" score, the customer is correctly classified as "Renew," making the model effective for decision-making despite the theoretical flaw.

Question 4: Integer Programming for Craft Cooperative

1. Integer Programming (IP) Model Formulation

1. Decision Variables:

Let ***L*** = Number of Bamboo Lamps to produce.

Let ***C*** = Number of Cane Baskets to produce.

Let ***J*** = Number of Jute Wall Hangings to produce.

2. Objective Function:

Maximize Total Profit ***Z***:

$$Maximize\ Z = 80L + 60C + 75J$$

3. Constraints:

- Raw Material Constraint: (Total available: 140 kg)

  $3L + 5C + 4J \leq 140$

- Artisan Time Constraint: (Total available: 90 hours)

  $2L + 1C + 3J \leq 90$

- Maximum Demand Constraint: (Max 20 of each)

  $L \leq 20$

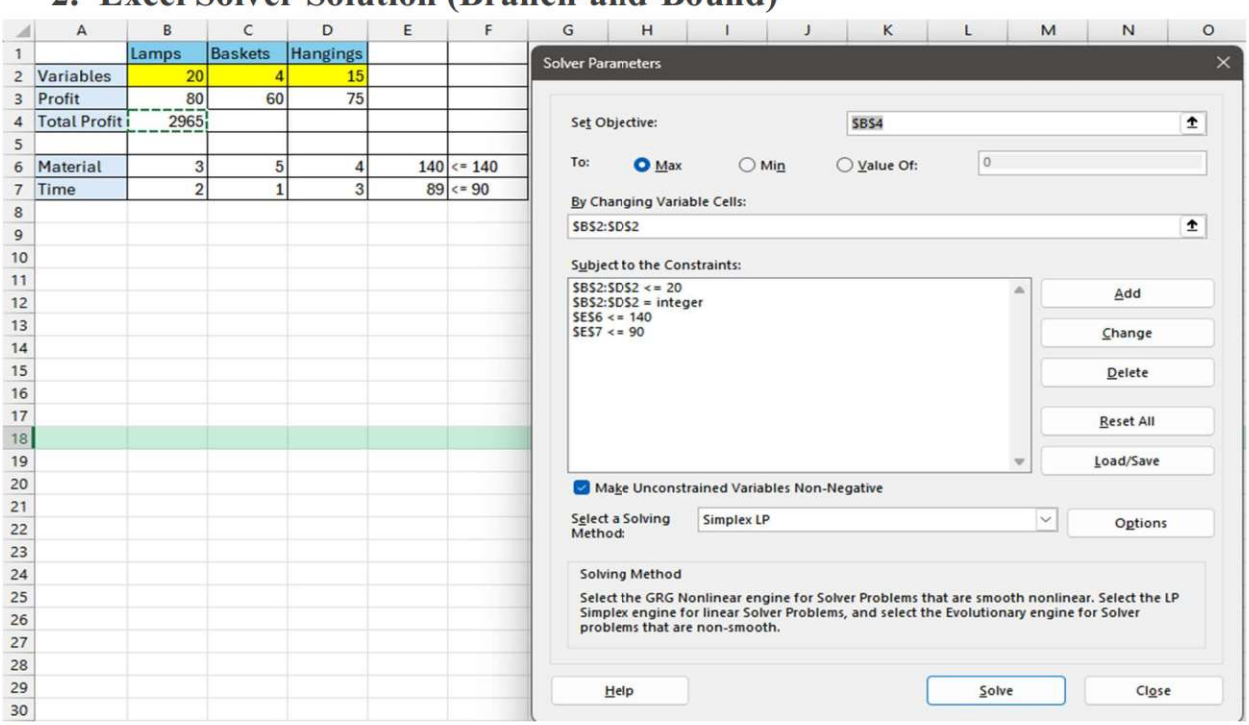
  $C \leq 20$

  $J \leq 20$

- Integer & Non-negativity Constraints:

  $L, C, J \geq$ and Integer

2. Excel Solver Solution (Branch-and-Bound)



	Lamps	Baskets	Hangings		
Variables	20	4	15		
Profit	80	60	75		
Total Profit	2965				
Material	3	5	4	140	<= 140
Time	2	1	3	89	<= 90

Microsoft Excel 16.0 Answer Report
Worksheet: [Book1]Sheet1
Report Created: 2025-12-02 9:22:51 PM
Result: Solver found a solution. All Constraints and optimality conditions are satisfied.
Solver Engine
Engine: Simplex LP
Solution Time: 0.047 Seconds.
Iterations: 1 Subproblems: 2
Solver Options
Max Time Unlimited, Iterations Unlimited, Precision 0.000001, Use Automatic Scaling
Max Subproblems Unlimited, Max Integer Sols Unlimited, Integer Tolerance 1%, Assume No

Objective Cell (Max)

Cell	Name	Original Value	Final Value
$B$4	Total Profit Lamps	0	2965

Variable Cells

Cell	Name	Original Value	Final Value	Integer
$B$2	Variables Lamps	0	20	Integer
$C$2	Variables Baskets	0	4	Integer
$D$2	Variables Hangings	0	15	Integer

Constraints

Cell	Name	Cell Value	Formula	Status	Slack
$E$6	Material	140	$E$6<=140	Binding	0
$E$7	Time	89	$E$7<=90	Not Binding	1
$B$2	Variables Lamps	20	$B$2<=20	Binding	0
$C$2	Variables Baskets	4	$C$2<=20	Not Binding	16
$D$2	Variables Hangings	15	$D$2<=20	Not Binding	5
$B$2:$D$2=Integer					

3. Optimal Production Plan & Financial Results

The Integer Programming model, solved using the Branch-and-Bound method (Simplex LP), yields the following optimal production plan:

- **Bamboo Lamps (L):** 20 units
- **Cane Baskets (C):** 4 units

- **Jute Wall Hangings (J):** 15 units

Total Maximum Profit: ₹2,965

Resource Usage:

- **Raw Material:** 140 kg used (100% of available 140 kg).
- **Artisan Time:** 89 hours used (Available 90 hours; 1 hour slack).

Question 5: Shortest Path Network Optimization for Logistics Route

1. Network Model Formulation

1. **Decision Variables:** Let *x_ij* be a binary variable (0 or 1) for each connection (arc) between location *i* and location *j*.

   - x_ij = 1 if the truck travels from *i* to *j*.
   - *x_ij* = 0 otherwise.

2. **Objective Function:** Minimize Total Transportation Cost:

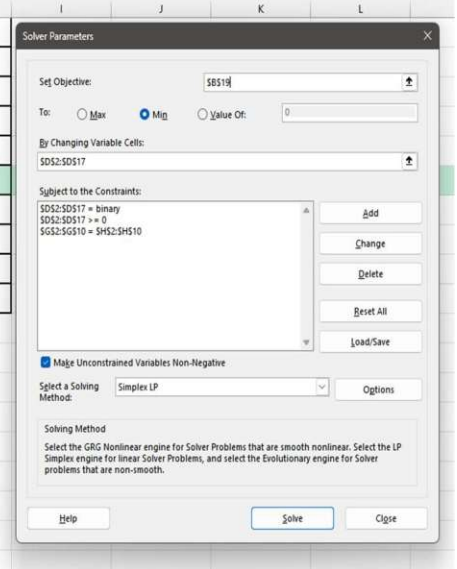
$$Minimize\ Z = \sum(\text{Cost}_{ij} \times x_{ij})$$

3. **Constraints (Flow Balance):** We must ensure the "flow" enters and leaves nodes correctly:
   - **Central Warehouse (Start):** Net Flow = +1 (1 unit leaves).
   - **RDC (End):** Net Flow = -1 (1 unit arrives).
   - **Intermediate Hubs (A, B, C...):** Net Flow = 0 (What enters must leave).

2. Excel Solver Solution

B19 =SUMIF($A$2:$A$17, F6, $D$2:$D$17) - SUMIF($B$2:$B$17, F6, $D$2:$D$17)

	A	B	C	D	E	F	G	H
1	Start	End	Cost	Flow (Leave empty/0)		Node	Net Flow Calculation	Required
2	CW	A	55	0		CW	0	1
3	CW	B	40	0		A	0	0
4	A	B	25	0		B	0	0
5	A	C	45	0		C	0	0
6	B	C	35	0		D	0	0
7	B	D	60	0		E	0	0
8	C	D	30	0		F	0	0
9	C	E	50	0		G	0	0
10	D	E	40	0		RDC	0	-1
11	D	F	70	0				
12	E	F	55	0				
13	E	G	65	0				
14	E	RDC	90	0				
15	F	G	40	0				
16	F	RDC	75	0				
17	G	RDC	50	0				
18								
19	Total Cost	0	=SUMPRODUCT(C2:C17, D2:D17)					



	A	B	C	D	E	F	G	H
1	Start	End	Cost	Flow (Leave empty/0)		Node	Net Flow Calculation	Required
2	CW	A	55	0		CW	1	1
3	CW	B	40	1		A	0	0
4	A	B	25	0		B	0	0
5	A	C	45	0		C	0	0
6	B	C	35	1		D	0	0
7	B	D	60	0		E	0	0
8	C	D	30	0		F	0	0
9	C	E	50	1		G	0	0
10	D	E	40	0		RDC	-1	-1
11	D	F	70	0				
12	E	F	55	0				
13	E	G	65	0				
14	E	RDC	90	1				
15	F	G	40	0				
16	F	RDC	75	0				
17	G	RDC	50	0				
18								
19	Total Cost	215						
20		=SUMPRODUCT(C2:C17, D2:D17)						

Microsoft Excel 16.0 Answer Report
Worksheet: [Book1]Sheet1
Report Created: 2025-12-02 10:02:12 PM
Result: Solver found a solution. All Constraints and optimality conditions are satisfied.
Solver Engine

Engine: Simplex LP
Solution Time: 0.063 Seconds.
Iterations: 10 Subproblems: 0

Solver Options

Max Time Unlimited, Iterations Unlimited, Precision 0.000001, Use Automatic Scaling
Max Subproblems Unlimited, Max Integer Sols Unlimited, Integer Tolerance 1%, Assume NonNegative

Objective Cell (Min)

Cell	Name	Original Value	Final Value
$B$19	Total Cost RDC	0	215

Variable Cells

Cell	Name	Original Value	Final Value	Integer
$D$2	A Flow (Leave empty/0)	0	0	Binary
$D$3	B Flow (Leave empty/0)	0	1	Binary
$D$4	B Flow (Leave empty/0)	0	0	Binary
$D$5	C Flow (Leave empty/0)	0	0	Binary
$D$6	C Flow (Leave empty/0)	0	1	Binary
$D$7	D Flow (Leave empty/0)	0	0	Binary
$D$8	D Flow (Leave empty/0)	0	0	Binary
$D$9	E Flow (Leave empty/0)	0	1	Binary
$D$10	E Flow (Leave empty/0)	0	0	Binary
$D$11	F Flow (Leave empty/0)	0	0	Binary
$D$12	F Flow (Leave empty/0)	0	0	Binary
$D$13	G Flow (Leave empty/0)	0	0	Binary
$D$14	RDC Flow (Leave empty/0)	0	1	Binary
$D$15	G Flow (Leave empty/0)	0	0	Binary
$D$16	RDC Flow (Leave empty/0)	0	0	Binary
$D$17	RDC Flow (Leave empty/0)	0	0	Binary

Constraints

Cell	Name	Cell Value	Formula	Status	Slack
$G$2	CW Net Flow Calculation	1	$G$2=$H$2	Binding	0
$G$3	A Net Flow Calculation	0	$G$3=$H$3	Binding	0
$G$4	B Net Flow Calculation	0	$G$4=$H$4	Binding	0
$G$5	C Net Flow Calculation	0	$G$5=$H$5	Binding	0
$G$6	D Net Flow Calculation	0	$G$6=$H$6	Binding	0
$G$7	E Net Flow Calculation	0	$G$7=$H$7	Binding	0
$G$8	F Net Flow Calculation	0	$G$8=$H$8	Binding	0
$G$9	G Net Flow Calculation	0	$G$9=$H$9	Binding	0
$G$10	RDC Net Flow Calculation	-1	$G$10=$H$10	Binding	0
$D$2	A Flow (Leave empty/0)	0	$D$2>=0	Binding	0
$D$3	B Flow (Leave empty/0)	1	$D$3>=0	Not Binding	1
$D$4	B Flow (Leave empty/0)	0	$D$4>=0	Binding	0
$D$5	C Flow (Leave empty/0)	0	$D$5>=0	Binding	0
$D$6	C Flow (Leave empty/0)	1	$D$6>=0	Not Binding	1
$D$7	D Flow (Leave empty/0)	0	$D$7>=0	Binding	0
$D$8	D Flow (Leave empty/0)	0	$D$8>=0	Binding	0
$D$9	E Flow (Leave empty/0)	1	$D$9>=0	Not Binding	1
$D$10	E Flow (Leave empty/0)	0	$D$10>=0	Binding	0
$D$11	F Flow (Leave empty/0)	0	$D$11>=0	Binding	0
$D$12	F Flow (Leave empty/0)	0	$D$12>=0	Binding	0
$D$13	G Flow (Leave empty/0)	0	$D$13>=0	Binding	0
$D$14	RDC Flow (Leave empty/0)	1	$D$14>=0	Not Binding	1
$D$15	G Flow (Leave empty/0)	0	$D$15>=0	Binding	0
$D$16	RDC Flow (Leave empty/0)	0	$D$16>=0	Binding	0
$D$17	RDC Flow (Leave empty/0)	0	$D$17>=0	Binding	0
$D$2:$D$17=Binary					

3. Optimal Route & Cost Analysis

Based on the network optimization model solved using Excel Solver, the minimum-cost route from the Central Warehouse to the Regional Distribution Center (RDC) is:

Optimal Route Sequence: Central Warehouse → Hub B → Hub C → Hub E → RDC

(Flow logic extracted from Solver output):

- **CW to B:** Flow = 1
- **B to C:** Flow = 1
- **C to E:** Flow = 1
- **E to RDC:** Flow = 1

Minimum Transportation Cost: ₹ 215